

Forensic Report - Fingerprint Comparison

Fingerprint Analysis Workstation - Version 2.3

Report generated at: 2026-05-08 14:55:15

Lab disclaimer: This report contains statistical software outputs and must not be used as sole identity evidence.

Audit Trail

Case Reference: -

Operator: -

SHA-256 (Reference file):
d61fa650ac6b127531ff88c5646682945450e3056cf255cec24737f356aba604

SHA-256 (Query file):
c02bdd597340fb459d97eafe38375c73c512854d55e277b6e10357ebb16286ac

Methodology Summary

Grayscale conversion, size normalization, CLAHE, optional denoising, Otsu/adaptive binarization, multi-orientation Gabor ridge enhancement, thinning, minutiae extraction (ending/bifurcation) with filtering, and one-to-one constrained matching by distance/angle/type. For partial-print verification, the query minutiae are searched with translation and small rotation inside the reference frame.

Recorded Run Parameters

Parameter	Value
border_margin	12
min_distance	14
min_contrast	20

min_angle_diff	12
denoise_method	fastNlMeans
fast_denoise_h	10
gauss_ksize	3
original_zoom	250
partial_zoom	73
partial_shift_x	0
partial_shift_y	0
apply_preview_scale	True
auto_scale_normalization	True
auto_scale_factor_applied	1.4
MATCH_DISTANCE_THRESHOLD	15
MATCH_ANGLE_THRESHOLD_DEG	35
PARTIAL_VERIFY_SEARCH_RADIUS	80
PARTIAL_VERIFY_STEP_PX	10

Pipeline Images

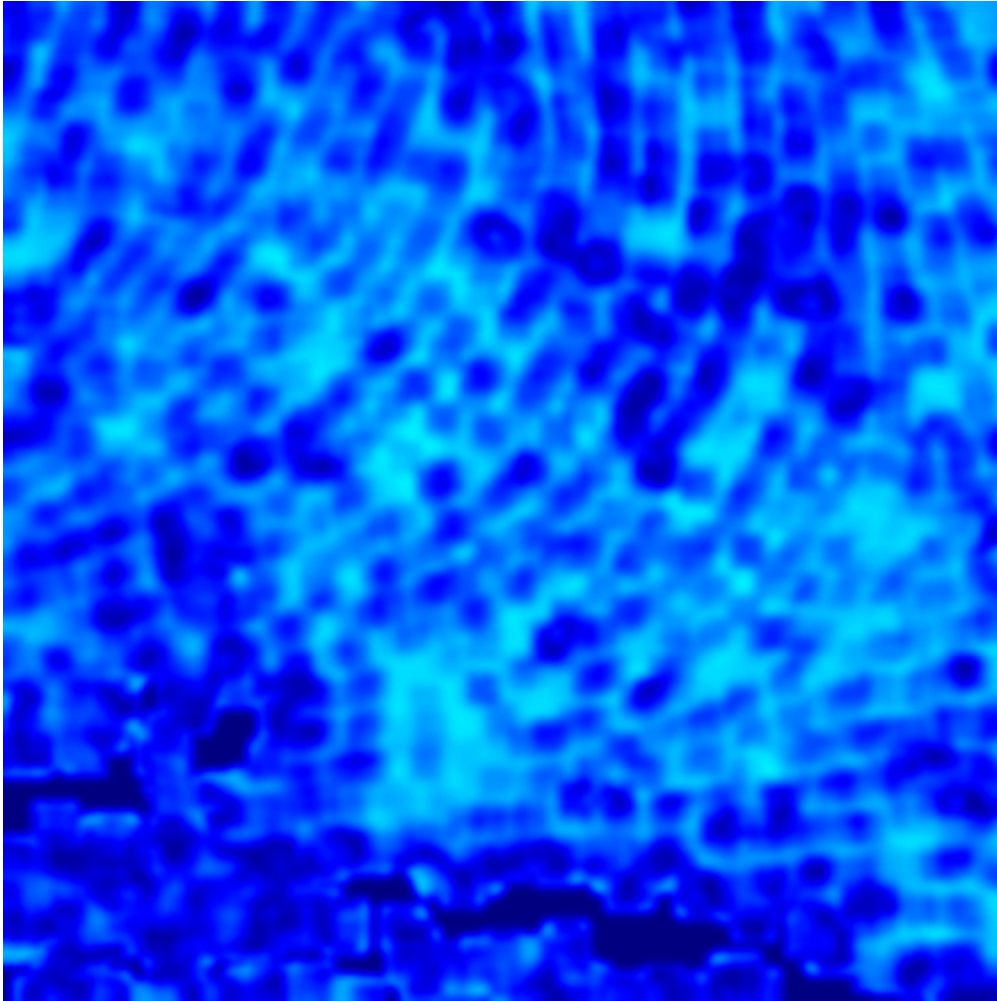
Reference Processing Pipeline

Stats: Binary white pixels: 118939 | Gabor white pixels: 43720 | Skeleton white pixels: 17569 | Filtered minutiae: 182

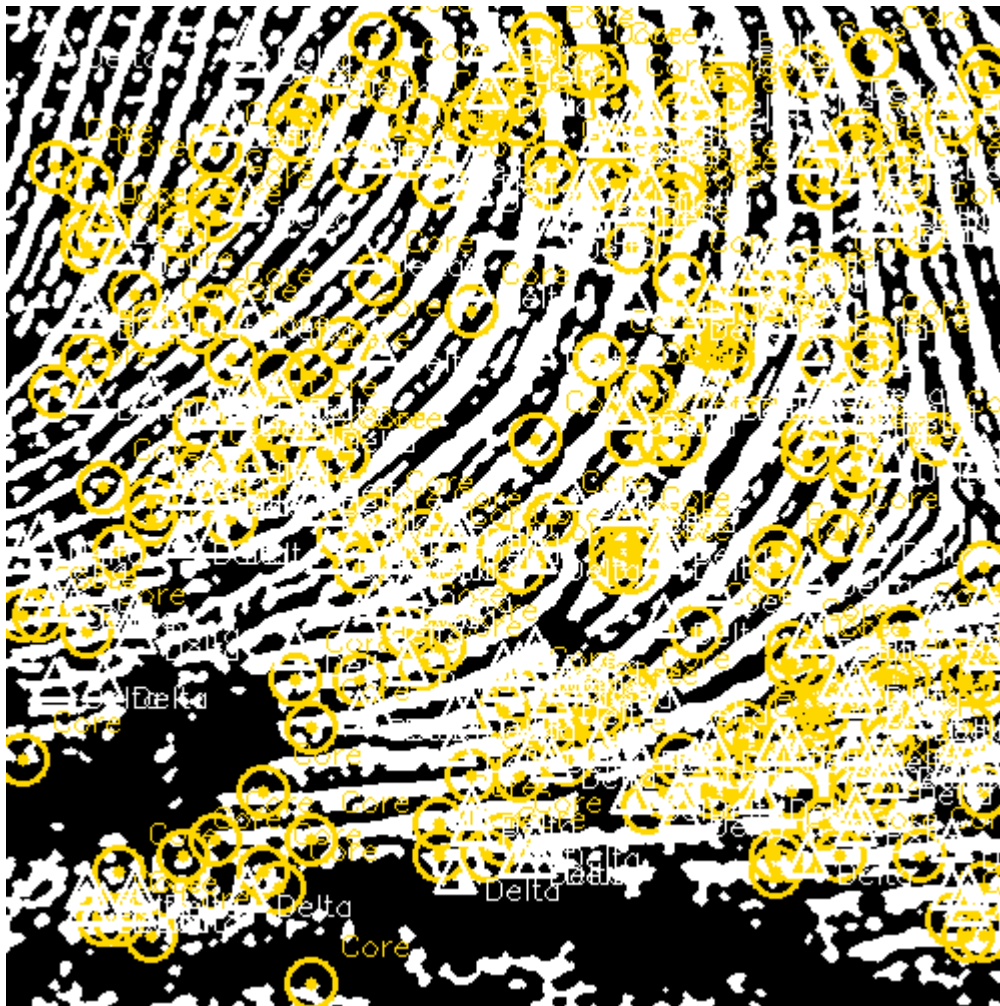
Preprocessed Binary



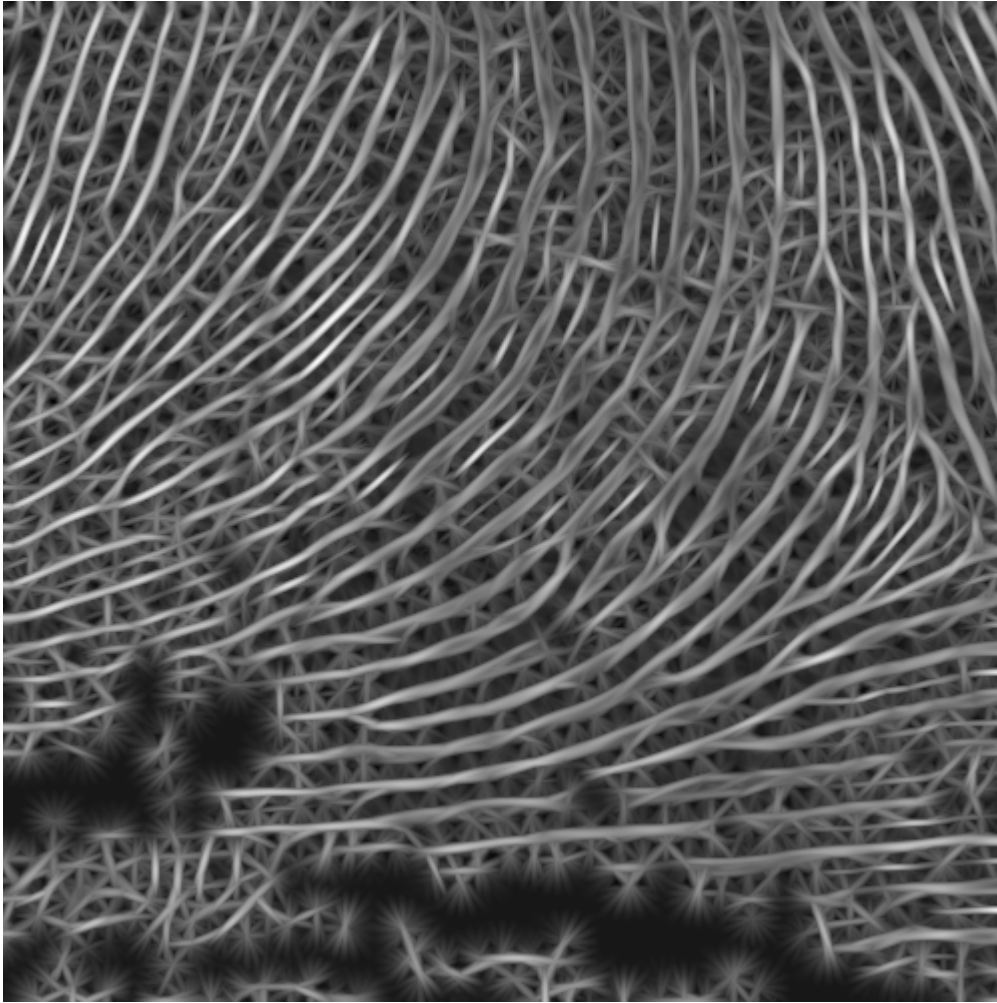
Quality Heatmap



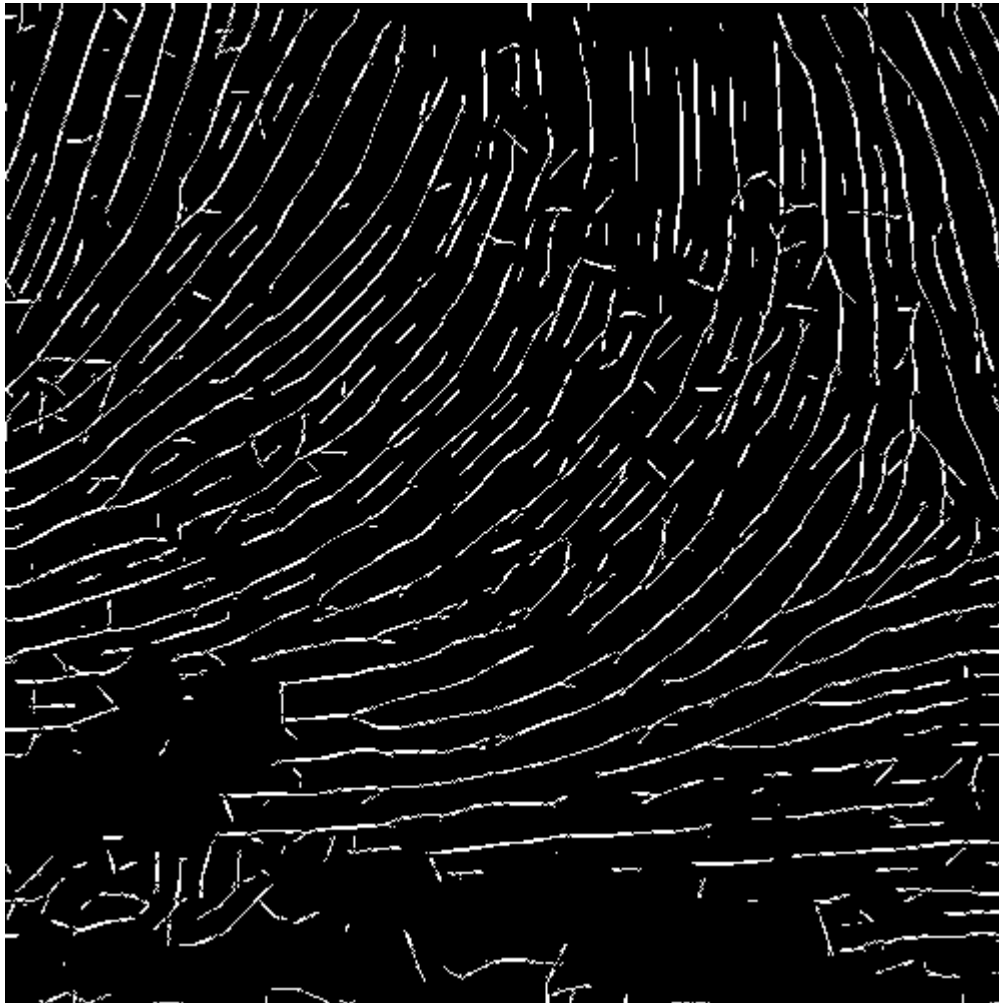
Singular Points



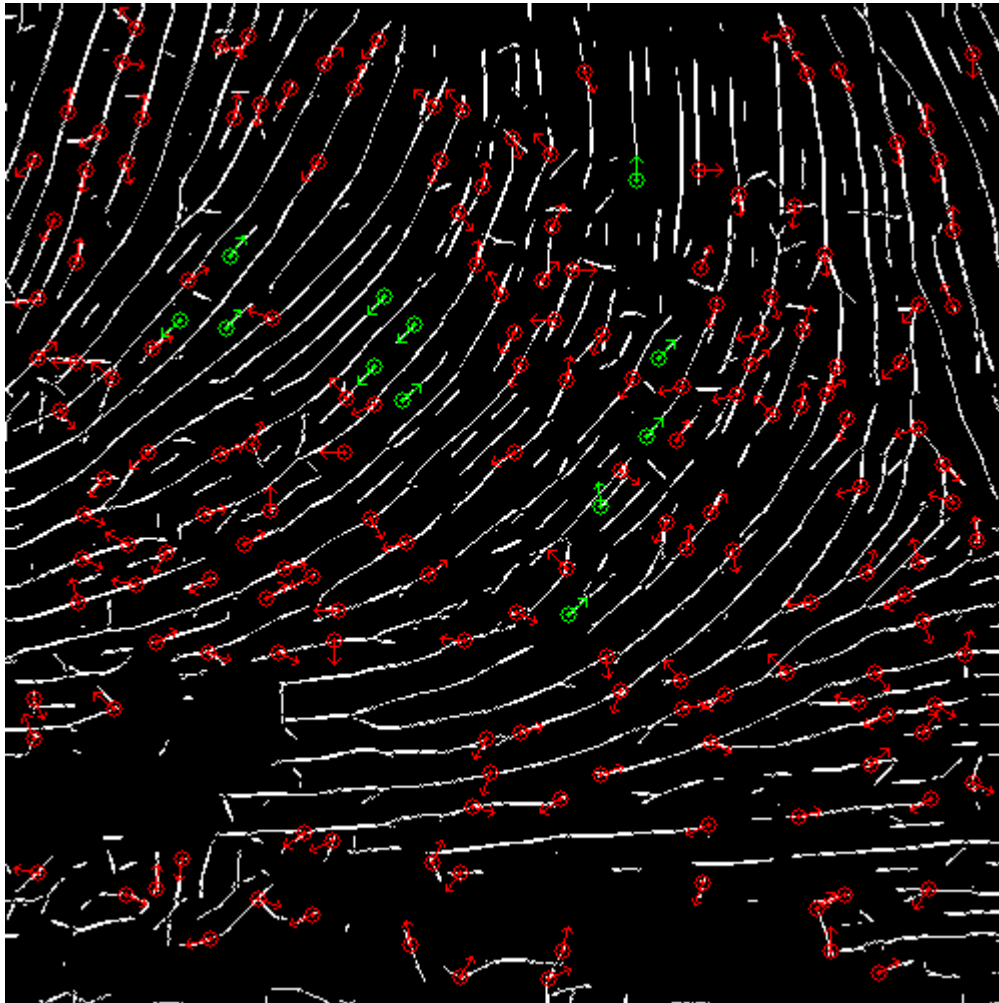
Ridge Enhancement (Gabor)



Skeleton



Minutiae Visualization



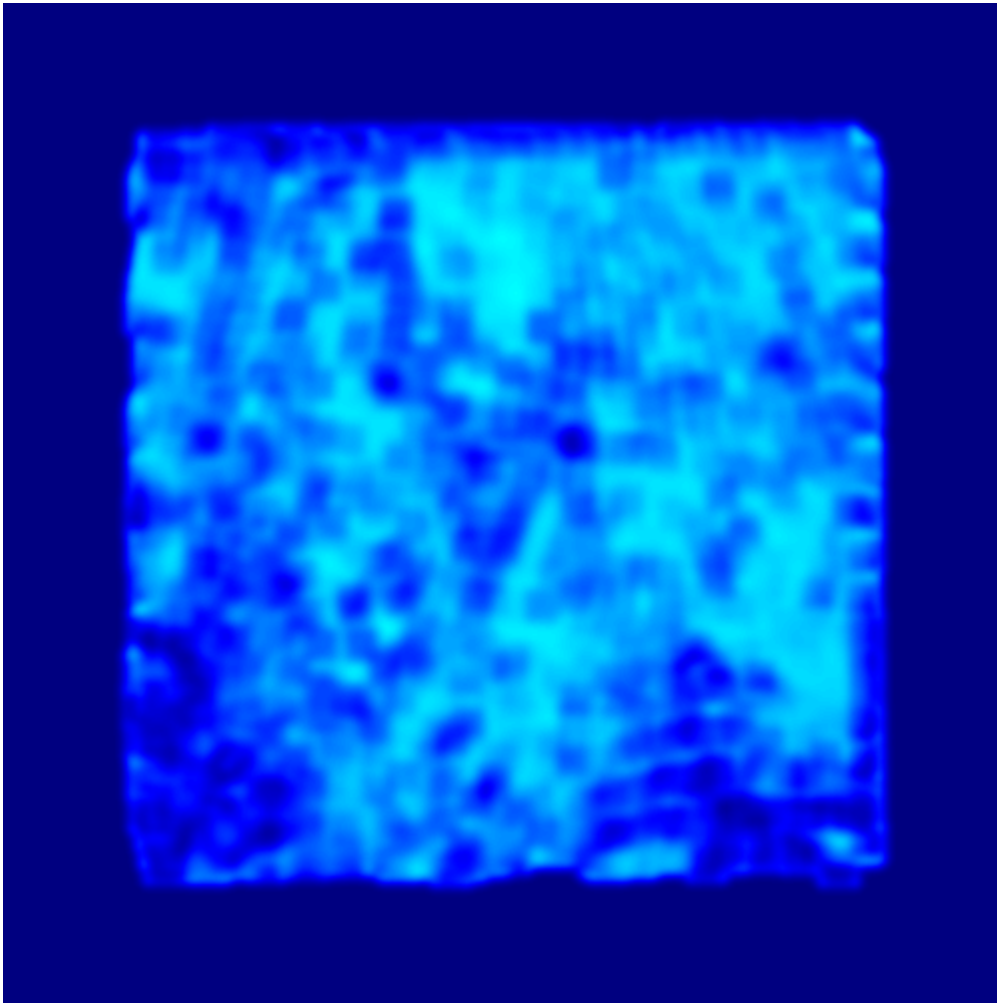
Query Processing Pipeline

Stats: Binary white pixels: 65856 | Gabor white pixels: 32631 | Skeleton white pixels: 11827 | Filtered minutiae: 108

Preprocessed Binary



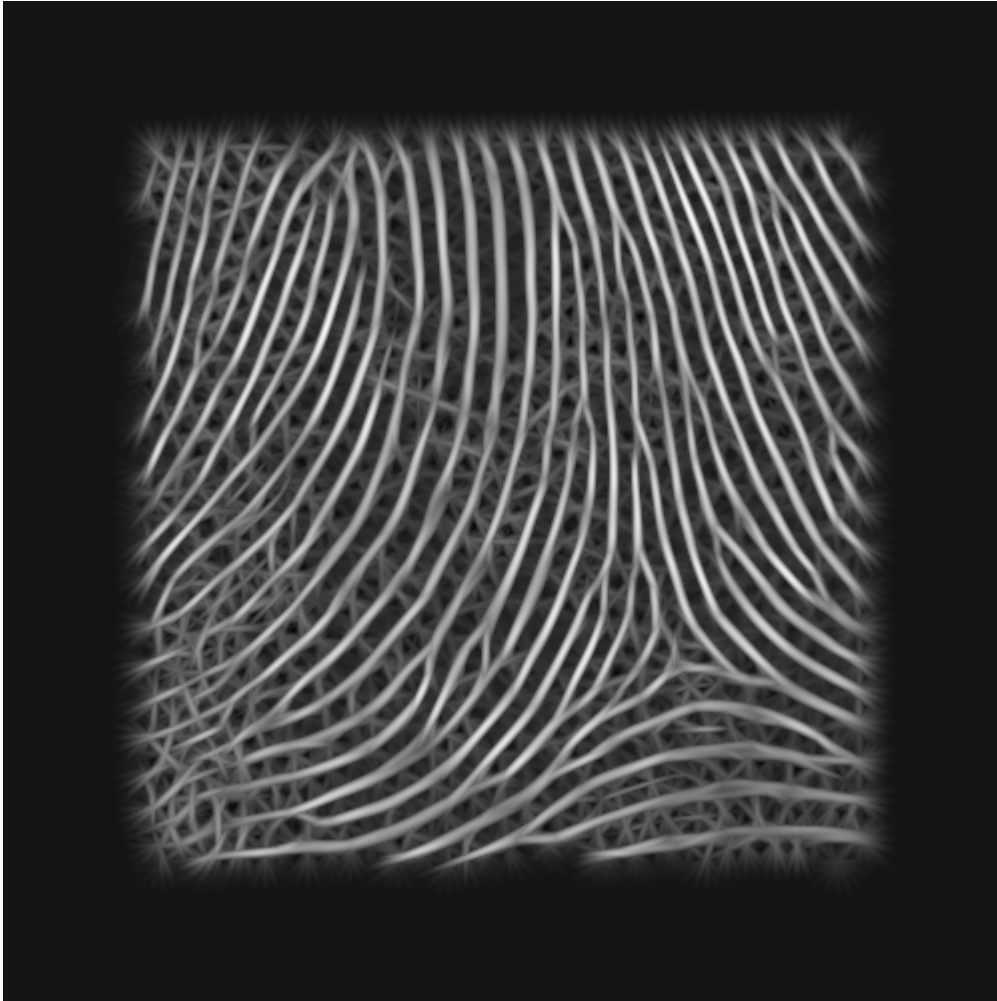
Quality Heatmap



Singular Points



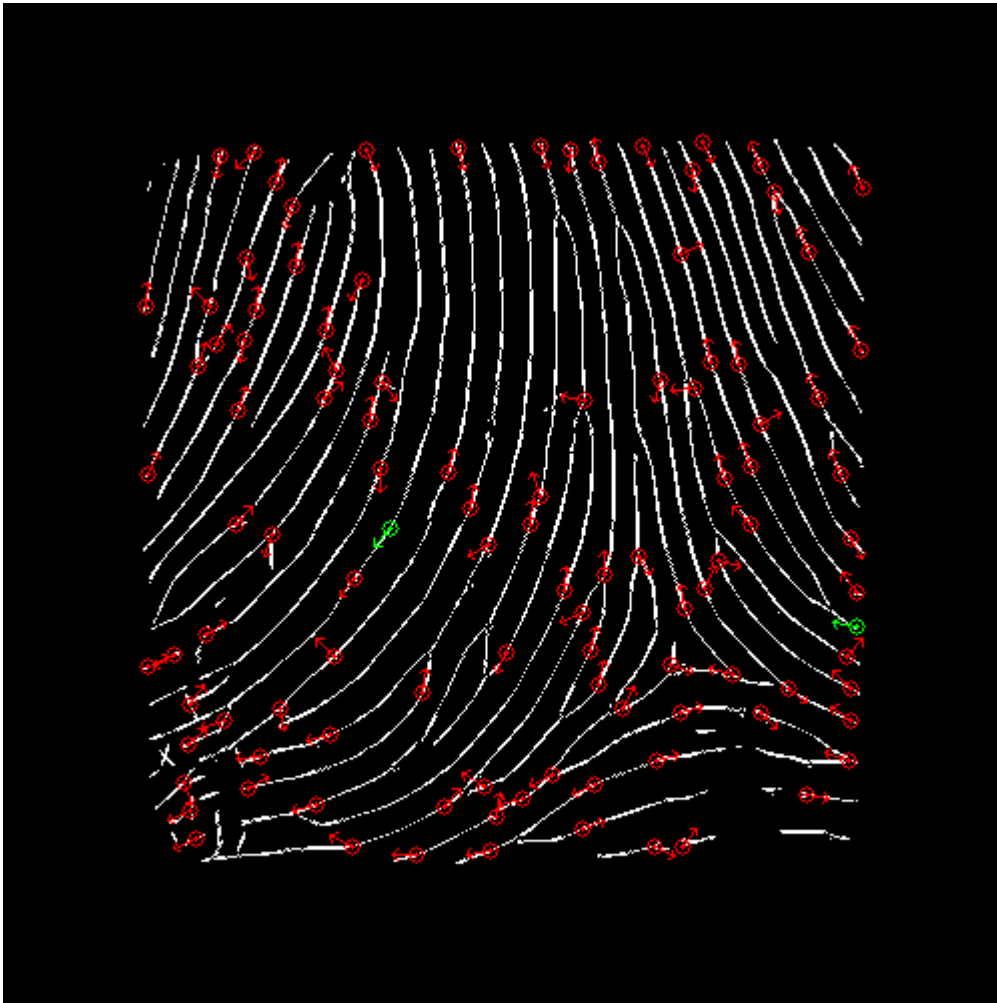
Ridge Enhancement (Gabor)



Skeleton

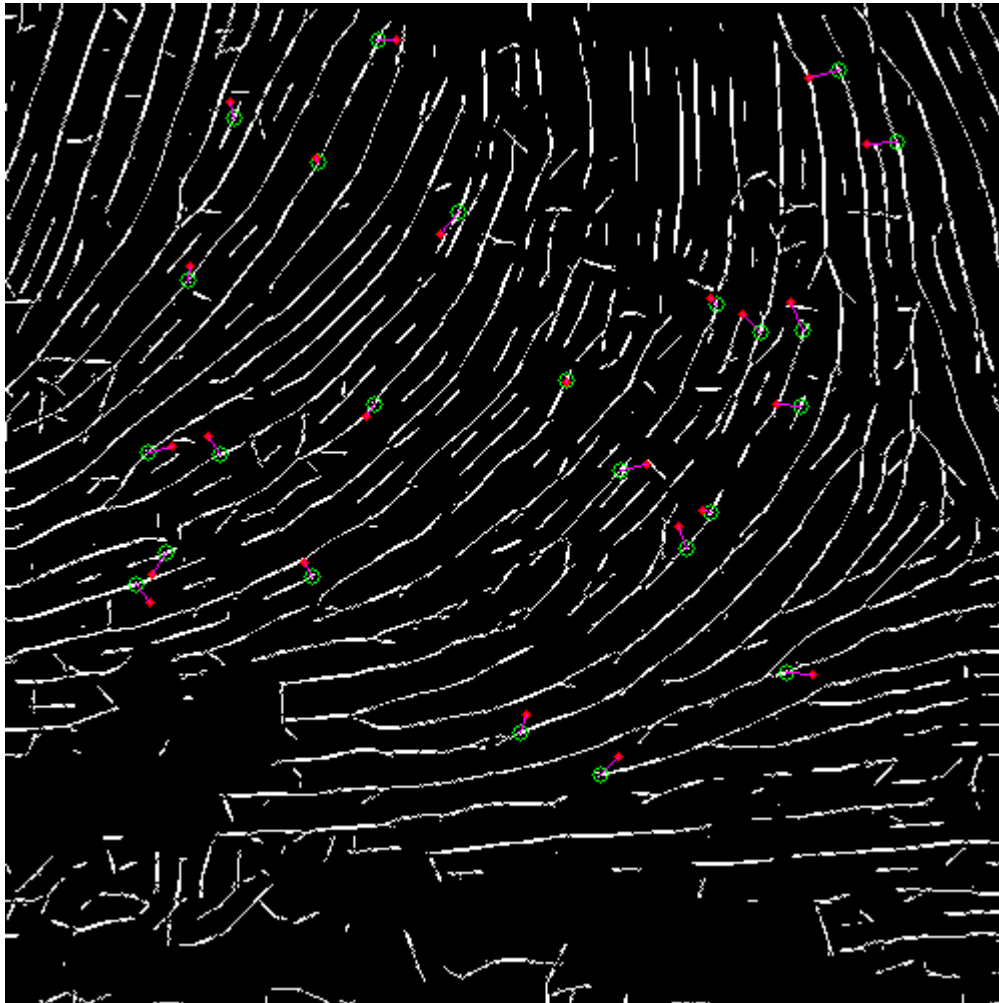


Minutiae Visualization



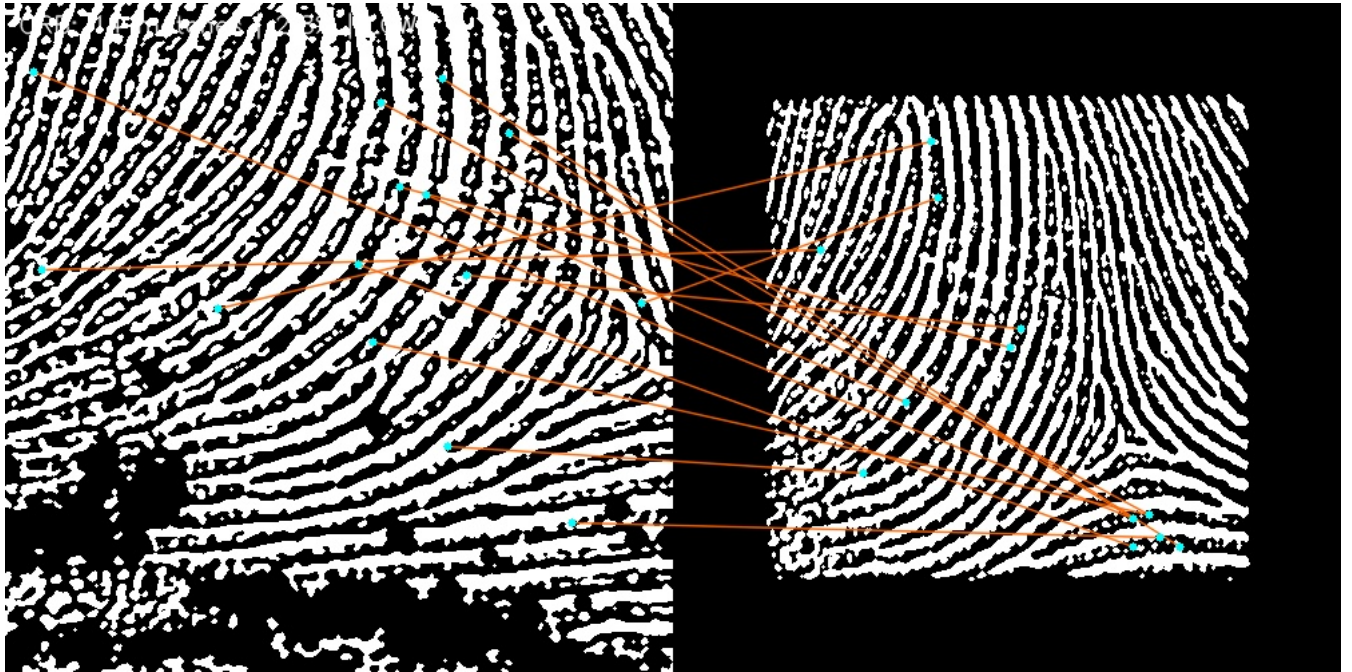
Aligned Match Visualization

Match Overlay



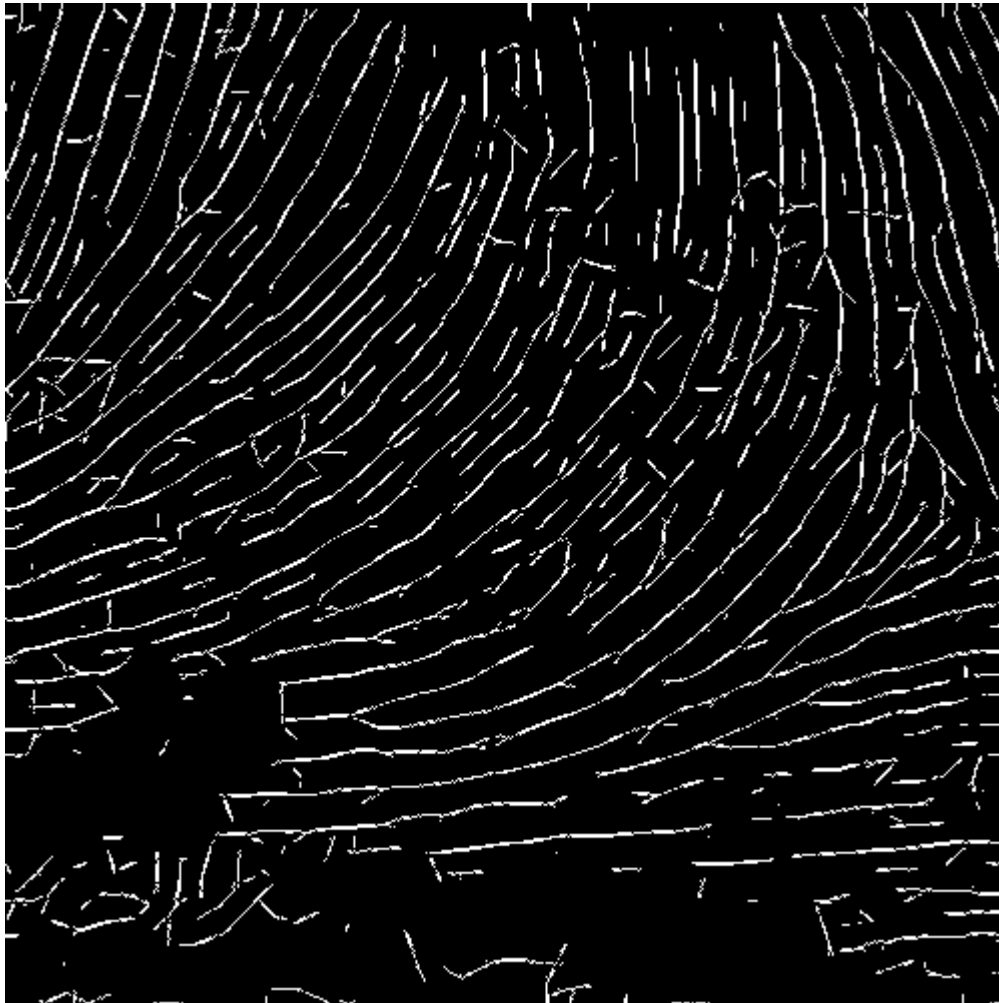
ORB Visual Verification

ORB Feature Matches



Final Skeletons Used for Matching

Reference Skeleton



Query Skeleton



Archive PNG files: `original_skeleton.png`, `partial_skeleton.png` in report folder.

Partial-to-Reference Alignment

Approximate alignment: $dx=12$ $dy=-61$ $rot=14.0$ deg. Baseline matches: 10, final matches: 24, gain: 14, score delta: +12.96.

Green: matched reference points | Orange: matched partial points after alignment.

Statistical Comparison Results

Reference minutiae count: 182

Query minutiae count: 108

One-to-one matched pairs: 24

Match score (query-based): 22.22%

Fused Score: 27.84%

Fusion Components - Minutiae: 22.22% | MCC: 47.43% | ORB: 2.80%

Score explanation: 24 / 108 = matched pairs divided by total extracted query minutiae. This is not area coverage.

Dice score: 16.55%

Forensic tier: Tier C - Low statistical similarity under configured thresholds

This system reports statistical metrics for fingerprint representation comparison under recorded parameters. It must not be used as sole identity proof and requires qualified expert review and lab SOP.

Internal status: LOW MATCH